

GDRV–HB half–bridge gate driver

Rev J0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Two NSI6651ASC channels for one leg of a CBB011M12GM4T module (DAB–D60)

01 Logic interface

J101 to the power board: +24V, 3.3 V, PWM_H/PWM_L (single-ended, already received from RS–422 on the power board), EN, FLT_N, RDY (pin 15 spare: GND)

File: sheets/01_interface.kicad_sch

02 High–side channel

NSI6651ASC + UCC14241–Q1 +15/–4 V, DESAT (3 x US1M), interlock with PWM_L, gate/Kelvin pins to the module

File: sheets/02_high.kicad_sch

03 Low–side channel

NSI6651ASC + UCC14241–Q1 +15/–4 V, DESAT (3 x US1M), interlock with PWM_H, gate/Kelvin pins to the module

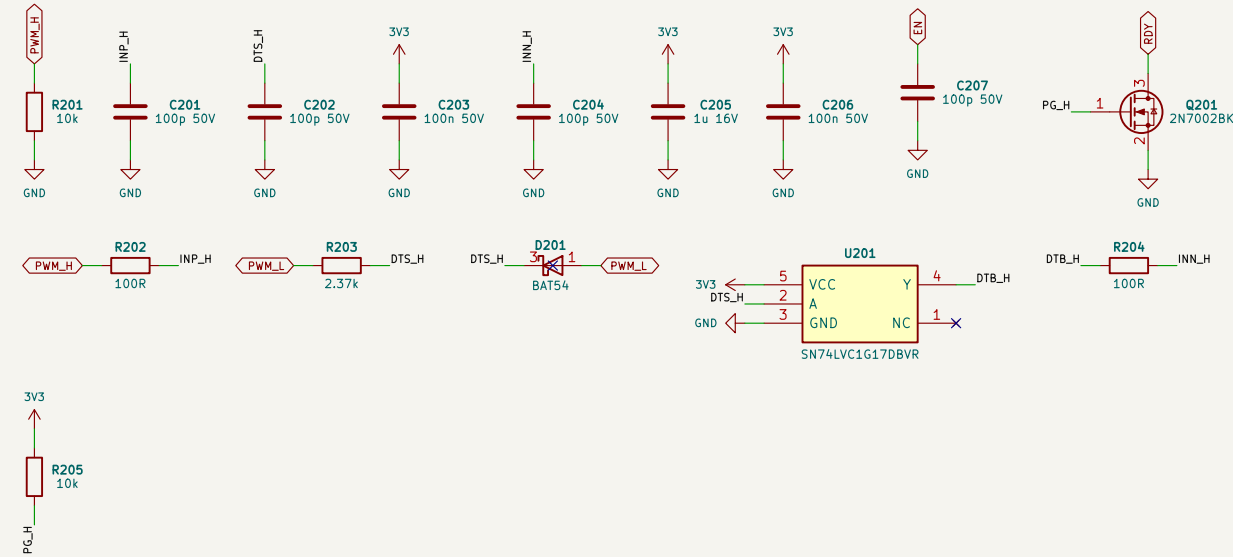
File: sheets/03_low.kicad_sch

CALCULATED by gen/gdrv.py design_check(), not measured (full list in outputs/GDRV–HB_design_check.txt). Gate rails +14.78..15.45 / –3.95..–4.07 V. Bias 1.02 W of 2.0 W for the module at 100 kHz; worst case 3 x C3M0016120K 1.53 W. Peak gate current 4.1 / 7.2 A (10 A rating). DESAT trips at V_DS 5.4–8.3 V after 263–810 ns blanking; worst–case gate–off 0.71 us (module) / 0.75 us (3 discretes) vs tSC 3 us of IMZA120R020M1H. Wolfspeed CBB011M12GM4T / C3M datasheets give no tSC: unverified. Hardware dead time 105–385 ns at the gates = max(firmware, hardware); firmware >= 315 ns is never stretched. Module NTC: plain analog line on the power board (no AIN). Barrier: NSI6651ASC VIOWM 2121 VDC (certified), UCC14241–Q1 1414 VDC (certification planned); CMTI >= 150 V/ns. Layout (not in this repo): card over the module signal pins, driver within 25 mm of J201/J301, gate loop < 100 mm (roadmap GDRV); D_H/D_L pins and DESAT strings need 1000 V + overshoot creepage per the ECO–10 insulation coordination.

Not bench–validated. PELV = logic side; HS, LS = floating channels; DCP = DC+ sense. Sheet 1 Schematics authored by Chinmoy Bhuyan Barrier: NSI6651ASC + UCC14241–Q1 reinforced (VIOWM 2121 / 1414 VDC) vs 1000 V bus		
Sheet: / File: GDRV–HB.kicad_sch		
Title: GDRV–HB half–bridge gate driver – Cover / Hierarchy		
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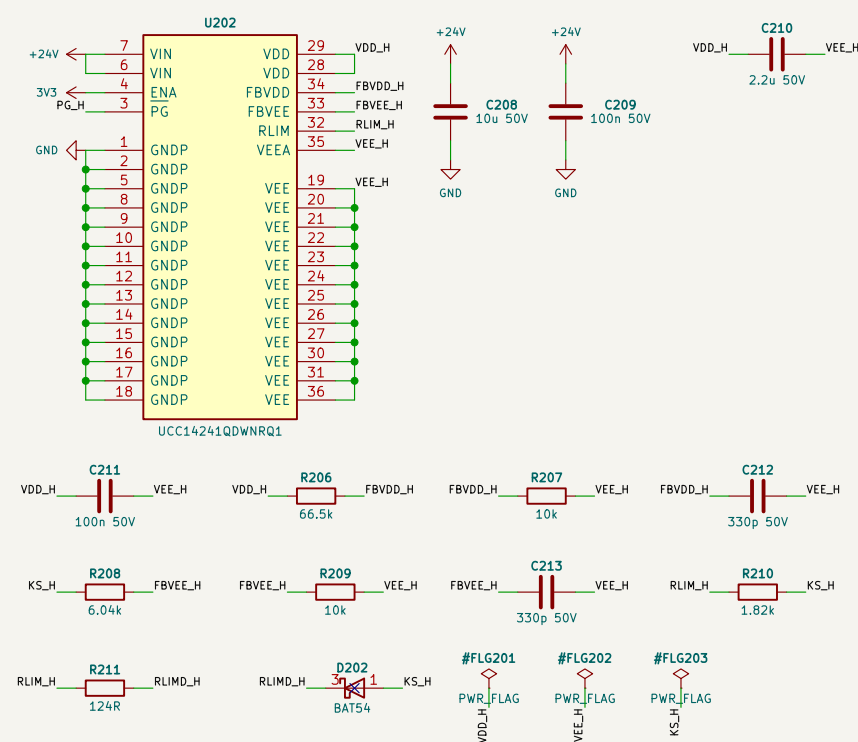


H: LOGIC SIDE



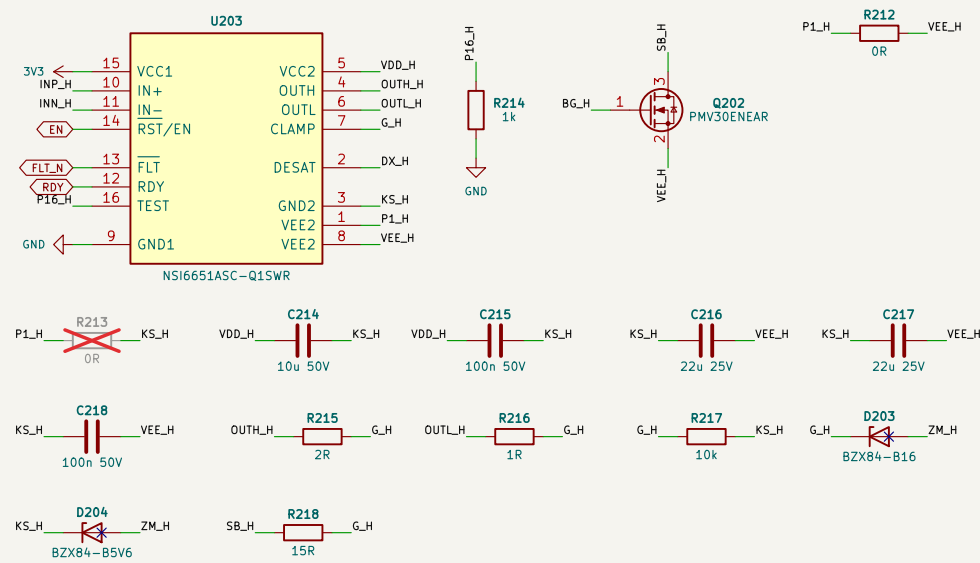
IN+ = own PWM; IN- = complementary switch, falling edge stretched 2.37k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

H: ISOLATED BIAS +15 V / -4 V



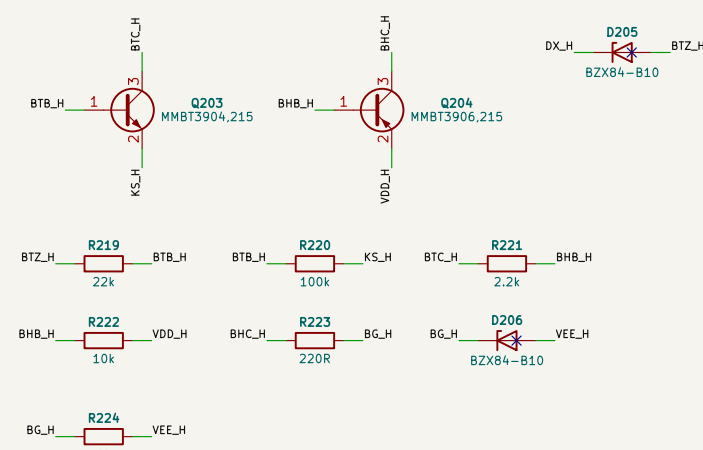
UCC14241-Q1, 24 V in, regulated +/-1.3 %, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

H: DRIVER AND GATE



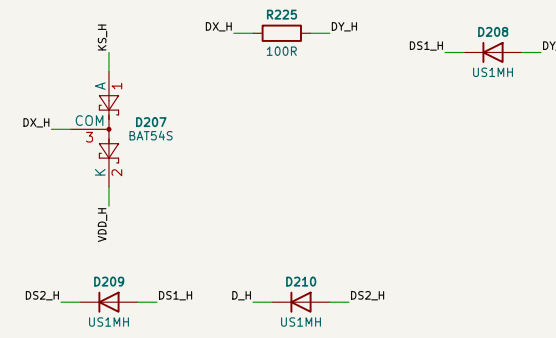
NSI6651ASC CLAMP direct to the gate (active Miller clamp). Kelvin = COM.
Zener pair clamps the gate to about +17 V / -6.6 V when the driver cannot.

H: SHORT-CIRCUIT BOOSTER



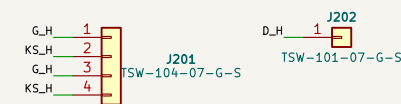
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 15R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

H: DESAT



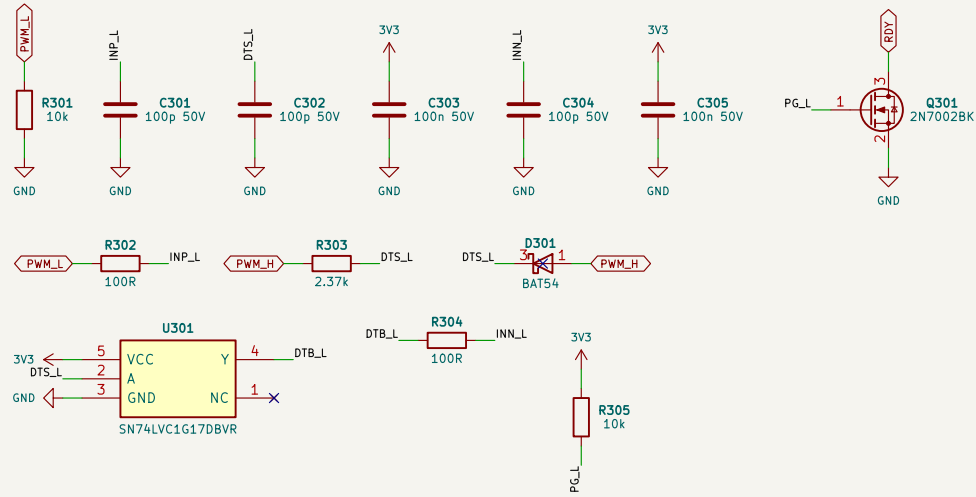
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

H: MODULE PINS



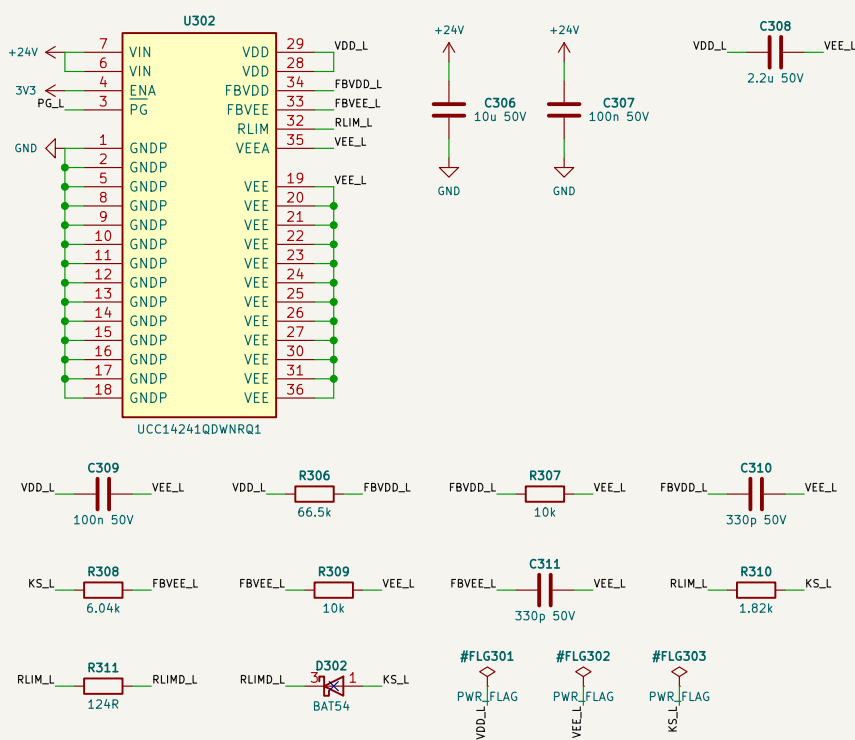
Solder pins to the power board under the module; G/KS interleaved. The module NTC is NOT on this card: plain analog line to the CELL AN2/AN3 on the power board. Drain sense on its own pin (module DC+), spaced for 1000 V.

L: LOGIC SIDE



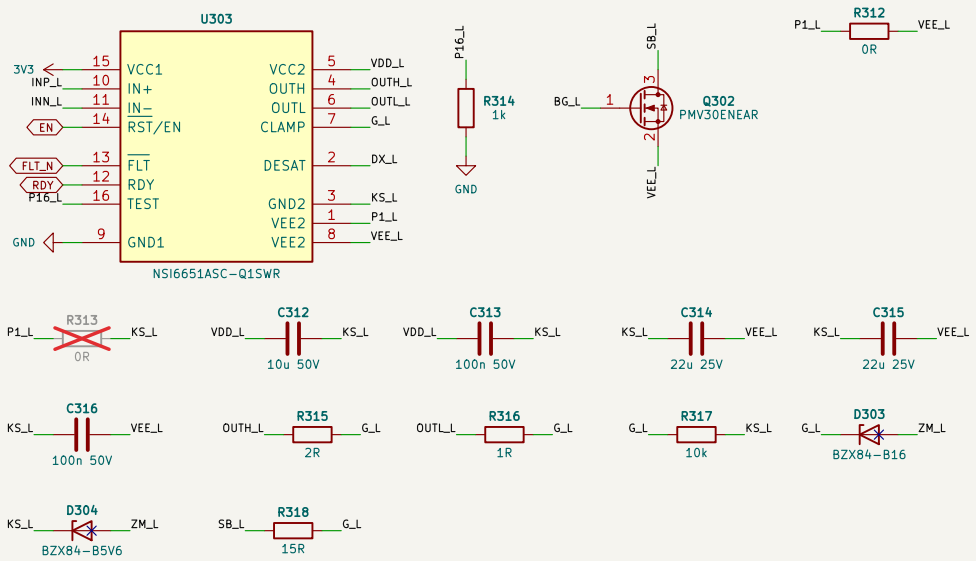
IN+ = own PWM; IN- = complementary switch, falling edge stretched 2.37k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

L: ISOLATED BIAS +15 V / -4 V



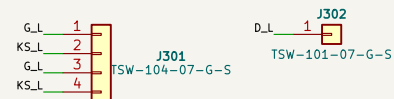
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

L: DRIVER AND GATE



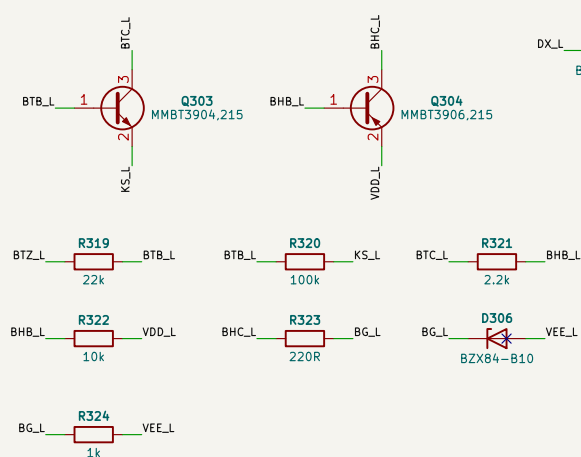
NSI6651ASC CLAMP direct to the gate (active Miller clamp). Kelvin = COM.
Zener pair clamps the gate to about +17 V / -6.6 V when the driver cannot.

L: MODULE PINS



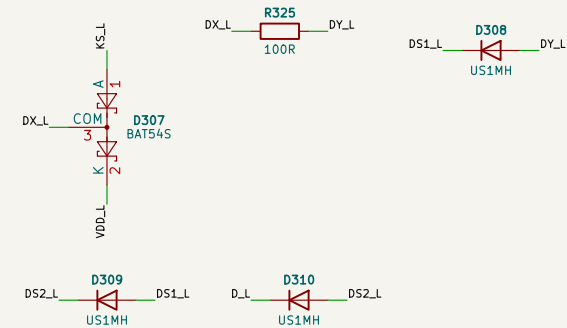
Solder pins to the power board under the module: G/KS interleaved. The module NTC is NOT on this card: plain analog line to the CELL AN2/AN3 on the power board. Drain sense on its own pin (module AC (switch node)), spaced for 1000 V.

L: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 15R pull the gate to VEE; BG holds -3 us (1k + FET gates).

L: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–9.8 V) – Rs – 3 x 50kΩ – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

Not bench-validated. PELV = logic side; HS, LS = floating channels; DCP = DC+ sense.
Sheet 4

Schematics authored by Chinmoy Bhuyan
Barrier: NSI6651ASC + UCC14241-Q1 reinforced (VIOWM 2121 / 1414 VDC) vs 1000 V bus

Sheet: /03 Low-side channel/
File: 03_low.kicad_sch

Title: GDRV-HB half-bridge gate driver – Low-side channel

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